Date: 18/03/2026

Duration: 1 hour 45

CONTINUOUS ASSESSMENT 1

II YEAR – SEMESTER 4

CSE23AE204 - PCP III

(B.Tech. E01,E02,E03,E05,E06, E61)

SET B

Question 1: Problem Statement

You are given a JSON object:

```
{
  "arr": [arrival_times],  "dep": [departure_times]
}
```

- arr[i] represents the arrival time of the ith train & dep[I] represents the departure time of the ith train.
- Arrival and departure times are given in 24-hour format (integer, e.g., 900 for 9:00 AM).

Rules:

- A platform cannot be shared if one train's arrival time is less than or equal to another train's departure time.
- If arrival time equals departure time, a new platform is required.

Your task is to determine the **minimum number of platforms required** so that no train waits.

Example 1

Input

"arr": [900, 940, 950, 1100, 1500, 1800], "dep": [910, 1200, 1120, 1130, 1900, 2000]

Output

3

Explanation

Arrival → 900, 940, 950, 1100, 1500, 1800 | Departure → 910, 1120, 1130, 1200, 1900, 2000

Overlap maximum occurs when:

- 900 arrives → 1 platform
- 940 arrives before 910 departs → 2 platforms
- 950 arrives before 910 departs → 3 platforms

Maximum platforms required = 3

Question 2: Problem Statement

An array of non-negative integers represents the **maximum jump length** that can be made from each index.

Starting at the **first index**, determine whether it is possible to reach the **last index** of the array.

From position i, a jump can reach any position in the range:

i + 1 to i + nums[i]

The task is to determine if reaching the final index is possible.

Example 1

Input

{"nums": [2,3,1,1,4]}

Output

true

Explanation

Possible jumps:

index 0 -> jump 2 -> reach index 2

index 2 -> jump 1 -> reach index 3

index 3 -> jump 1 -> reach index 4

Last index is reachable.

Input Format

{"nums": [a1, a2, a3, ...]}

Where:

- nums[i] represents the **maximum jump length** from index i.

Output Format

Return:

true

or

false

Where:

- true → last index is reachable
- false → last index cannot be reached

Question 3: Problem Statement

A logistics company needs to load packages onto a truck for delivery. The truck has a fixed **maximum weight capacity**.

You are given a list of packages:

packages = [[weight1, value1], [weight2, value2], ...]

Each package contains:

- **weight** → weight of the package
- **value** → value of the package

Unlike traditional problems, **packages can be partially loaded**.

If only a fraction of a package is loaded, the value gained is proportional to the fraction of the weight taken.

For example:

- A package with weight = 10 and value = 100
- If 5 units are loaded → value gained = 50

Your task is to determine the **maximum total value** that can be loaded onto the truck without exceeding its capacity.

Example 1

Input

packages = [[10,60],[20,100],[30,120]]

capacity = 50

Output

240

Explanation

Steps:

- Take (10,60) fully → 60
- Take (20,100) fully → 100
- Remaining capacity = 20
- Take 20/30 of last → 80

Total = **240**